

Design and Development of a Real Estate CRM with Revenue Sharing Management System

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Abstract—The real estate industry involves complex business activities including property management, customer relationship management, lead tracking, sales coordination, and multi-stakeholder commission distribution. At present, the majority of real estate enterprises rely on manual record-keeping, disjointed spreadsheets, and isolated software tools, resulting in inconsistent customer data, communication barriers, commission disputes, and a severe 65% to 85% industry CRM implementation failure rate. This article describes the design, implementation, and empirical validation of EstateFlow CRM, an enterprise-grade web-based platform that unifies property listings, customer lifecycle tracking, behavioral lead scoring, and an automated Revenue Sharing Management System into a cohesive architecture. Engineered with an asynchronous FastAPI (Python) backend and a reactive React 19 + TypeScript frontend backed by a self-healing PostgreSQL/SQLite dual-engine persistence layer, the system automates multi-tier commission calculations upon booking confirmation, enforces statutory 5% Tax Deducted at Source (TDS under Section 194H), and maintains immutable double-entry wallet transaction ledgers. Furthermore, it incorporates a 5-factor conversion-weighted lead-scoring heuristic driving a 6-stage Kanban pipeline, together with builder RERA compliance auditing and mortgage EMI calculators. Extensive staging evaluation confirms zero TypeScript compilation defects across 2,530 modules, sub-60ms P99 API latency across all routes, and a 100% pass rate across 16 automated end-to-end integration test suites.

Keywords—Real Estate, Customer Relationship Management, Revenue Sharing, Commission Management, PropTech, FastAPI, React 19, Statutory TDS Withholding, Role-Based Access Control (RBAC).

I. INTRODUCTION

The real estate industry plays a foundational role in driving global economic growth and capital formation. Core commercial activities encompass property development, brokerage representation, client relationship management, and financial transaction settlement [1]. As competition intensifies, real estate brokerages and property developers require modern digital tools to streamline operations and enhance client satisfaction. A pivotal instrument in this transformation is Customer Relationship Management (CRM) software, designed to centralize client interactions, property inventories, and transaction pipelines [1], [2].

Historically, real estate organizations have relied on manual ledger entries, disparate spreadsheet workbooks, or disconnected commercial software to manage property inventories, leads, and sales activities [3]. This traditional operating model produces severe vulnerabilities:

- **Data Inconsistency & Silos:** Customer inquiries, site visits, and purchase histories remain fragmented across individual sales agents' personal records, resulting in severe data loss during staff turnover [1], [5].
- **Commission Friction & Delayed Settlements:** Real estate commissions involve multi-tier compensation models (percentage splits, flat fees, performance slabs). Manual reconciliation results in human calculation errors, accounting discrepancies, and payment delays exceeding 60 to 90 days, causing acute distrust between brokers and channel partners [1], [7].
- **Statutory Tax Non-Compliance:** Brokerage transactions are subject to strict statutory fiscal mandates. In India, Section 194H of the Income Tax Act mandates 5% Tax Deducted at Source (TDS) withholding on commission disbursements, alongside Real Estate Regulatory Authority

(RERA) builder verification. Generic CRMs lack native tax withholding triggers [1].

- **Alarming Implementation Failure Rates:** Empirical research by Ferreira et al. [1] reveals that 65% to 85% of real estate CRM implementations fail due to software over-complexity, lack of user-centered design, and excessive administrative overhead.

By integrating CRM capabilities and automated revenue sharing into a unified, lightweight cloud architecture, real estate enterprises can replace manual tracking with an auditable, automated workflow.

This research project designs, implements, and evaluates **EstateFlow CRM**—a web-based Real Estate CRM with an integrated Revenue Sharing Management System. The specific research objectives are:

1. Develop a centralized platform for property listings, leads, customers, and sales operations.
2. Engineer an automated revenue-sharing engine that calculates tiered commissions and enforces statutory 5% TDS withholding.
3. Establish immutable double-entry wallet transaction ledgers to guarantee financial auditability.
4. Deploy a conversion-weighted lead scoring heuristic ($SS_L \in [0, 100]\%$) and 6-stage Kanban sales pipeline.
5. Empirically benchmark system build stability, API latencies, and end-to-end integration correctness.

II. LITERATURE REVIEW & RELATED WORK

The development of EstateFlow CRM synthesizes insights across real estate CRM design, digital transformation frameworks, algorithmic recommendation systems, and revenue-sharing contract theory.

A. Real Estate CRM Design & User Experience

Ferreira et al. [1] investigated why enterprise CRMs exhibit 65% to 85% failure rates in real estate. Applying Design Science Research (DSR) and the Stanford/IDEO Design Thinking 5-phase model across five Portuguese real estate agencies and nine expert-review cycles, they demonstrated that user adoption depends on six prioritized web modules: Contacts, Client Profile & Qualification, Calendar & Agenda Integration, Dashboard & Goals, Business Funnels (Pipelines), and Notifications [1]. They highlighted that applying cognitive UX laws—such as *Occam's Razor* (utility before aesthetics), the *Aesthetic-Usability Effect*, *Gestalt grouping*, and *Tesler's Law of Conservation of Complexity*—substantially improves task success and user satisfaction [1].

B. CRM Adoption & Organizational Restrictive Factors

Tien et al. [2] analyzed CRM adoption at Gamuda Land's Celadon City project in Vietnam, identifying data integration, staff training, and executive support as determinants of CRM success. Complementing this, Korsakienė, Tvaronavičius, and Mačiulis [9] conducted an empirical survey of 90 real estate firms, finding that while 98% of firms agree CRM elevates managerial competence and 100% of domestic firms report sales expansion, adoption is impeded by organizational culture resistance (97%), inadequate leadership (91%), and high employee turnover (70%). These findings justify EstateFlow's focus on centralized data custody and intuitive workflows.

C. Digital Transformation Technologies in Real Estate

Al-haimi et al. [3] performed a systematic PRISMA literature review of 36 empirical studies, establishing a 3-dimensional conceptual framework: Technologies (AI, IoT, blockchain, AR/VR, BIM), Benefits (operational efficiency, transparency, cost savings), and Challenges (high implementation cost, security apprehensions, regulatory compliance). Al-haimi et al. specifically advocated for *modular, lightweight technology platforms* that optimize financial outlays for small and medium-sized real estate enterprises [3].

D. Property Recommendations & Production Latency SLAs

Gharahighehi, Pliakos, and Vens [4] surveyed 26 real estate recommendation studies, identifying core domain challenges: cold-start problems, domain-specific spatial features, conflicting buyer criteria, and extreme data sparsity. Venkatesh et al. [5] implemented *RE-RecSys* for Housing.com in India (3 million monthly users), establishing a 4-tier user classification (cold-start, short-term, long-term, hybrid) and an empirical conversion-weighted action hierarchy: CRF submission (weight = 10), OTP verification (weight = 8), and detailed page views (weight = 4), under a strict production latency SLA of <40 ms [5].

E. Revenue-Sharing & Coordination Contract Theory

Lan and Yu [7] formulated a decentralized revenue-sharing–commission coordination contract for multi-stakeholder supply chains, proving that calibrated sharing rates align independent agent incentives and maximize joint profits. De Giovanni and Roselli [8] demonstrated that pure percentage splits risk channel instability unless supported by structured rules and dispute-mitigating safeguards. These models provide the theoretical foundation for EstateFlow's configurable commission engine.

TABLE I: SYSTEMATIC SUMMARY OF REVIEWED LITERATURE

Ref	Focus Area	Key Academic Contribution	Relevance to EstateFlow
[1]	CRM UX / DSR	Identifies 65-85% failure rate; establishes 6 core modules and UX psychology	Direct blueprint for EstateFlow's 12-module UI/UX console
[2]	CRM Adoption	Analyzes Gamuda Land; emphasizes staff training and data accessibility	Informs lead tracking and centralized client histories
[3]	PropTech SLR	PRISMA review of 36 studies; 3D tech-benefit-challenge framework	Justifies lightweight cloud architecture and security controls
[4]	RecSys Survey	Taxonomy of CF/CB models; cold-start and conflicting criteria hurdles	Guides parametric search and multi-attribute property compare
[5]	RE-RecSys	4-tier user hierarchy, conversion-weighted feedback, <40ms latency SLA	Calibrates EstateFlow's lead scoring and API response latency
[6]	AI Marketing	Automated lead capture, chatbot engagement, and digital workflows	Supports automated lead ingestion and notification dispatches
[7]	Revenue Contracts	Decentralized RS-CC coordination contract maximizing channel profit	Mathematical foundation for tiered commission rule engine
[8]	Contract Stability	Demonstrates need for contract support programs to ensure fairness	Justifies commission caps, dispute guards, and audit logs

III. SYSTEM ARCHITECTURE & METHODOLOGY

A. Development Methodology

EstateFlow CRM was engineered using an incremental, module-by-module Software Development Life Cycle (SDLC). Each functional subsystem—Administrator Authentication, Admin Dashboard, Property Management, Builder CRM, Revenue Engine, and Lead Scoring—was implemented as an independent step verified with dedicated automated test harnesses prior to subsequent integration.

B. Three-Tier Client-Server Architecture

The platform follows a decoupled, three-tier cloud architecture separating presentation, business logic, and data persistence:

- **Presentation Layer (Client):** Implemented in **React 19** bundled with **Vite 8** and styled via **Tailwind CSS 3.4**. Server-state caching and optimistic UI updates are managed by **TanStack React Query v5**. Administrative route-guarding is enforced using **react-router-dom v7** via an **AdminProtectedRoute** wrapper inspecting decoded JWT claims. Visualizations (revenue trends, lead funnels, EMI amortization) are rendered with **Recharts**, while geospatial mapping is powered by **Leaflet**.
- **Application Logic Layer (Server):** Constructed on **FastAPI** running Python 3.11+ over the **Uvicorn ASGI** server (Starlette engine). It enforces clean separation of concerns: **config/** (settings), **core/** (security, JWT, RBAC), **routers/** (REST endpoints), **schemas/** (Pydantic v2 validation), **services/** (revenue and lead engines), and **repositories/** (isolated database operations).
- **Data Persistence Layer:** Modeled with **SQLAlchemy 2.0 ORM** across a self-healing dual-engine configuration: production environments connect to enterprise **PostgreSQL 15+** (pool size 10, overflow 20, pre-ping enabled), while local development and staging seamlessly fail over to an embedded **SQLite** database (**estateflow.db**) with thread-safety guards.

TABLE II: TECHNOLOGY STACK USED IN ESTATEFLOW CRM

Layer	Technology	Architectural Purpose
Frontend	React 19 + Vite 8	Component SPA with fast builds and reactive DOM
Styling	Tailwind CSS 3.4	Utility-first responsive styling with dark/light tokens
State Sync	TanStack Query v5	Client-side caching, revalidation, optimistic mutations
Client Routing	react-router-dom v7	Role-based routing with JWT claims inspection
Charts & Maps	Recharts & Leaflet	Financial revenue curves, funnels, and GIS mapping
Backend API	FastAPI (Py 3.11+)	Asynchronous ASGI REST API with auto OpenAPI docs
App Server	Uvicorn Starlette	Non-blocking concurrent event loop request handling
ORM Engine	SQLAlchemy 2.0	Dual-engine declarative relational persistence
Primary DB	PostgreSQL 15+	Production data store with connection pooling
Fallback DB	SQLite (embedded)	Automatic local/staging fallback (estateflow.db)
Validation	Pydantic v2	Strict JSON schema serialization and validation
Cryptography	passlib (bcrypt)	Salted one-way hashing of user passwords
Tokens	JWT (HS256)	30-minute access tokens, 7-day sliding refresh
Containers	Docker Compose	Multi-stage containerized cloud deployment

IV. SYSTEM DESIGN & CORE WORKFLOWS

A. Context-Level Data Flow (DFD Level 0)

EstateFlow CRM interacts with three primary external entities: Administrative/Sales Staff, Retail Customers (Buyers), and Builders/Channel Partners. The system receives listing data, booking requests, and revenue configurations, while dispatching verified listings, commission credits, monthly settlement statements, and in-app notifications.

B. Entity-Relationship Data Architecture

The database schema is centered around a core **Booking** entity bridging **User** and **Property** records, linked to a **Lead** entity capturing behavioral scoring, and a **RevenueRule** / **CommissionRecord** pair driving automated disbursements into **Wallet** and **WalletTransaction** audit ledgers.

C. Role-Based Access Control (RBAC)

To protect commercial data and prevent unauthorized commission manipulation, EstateFlow enforces a 12-permission matrix across six distinct user roles.

TABLE III: ROLE-BASED ACCESS CONTROL (RBAC) PERMISSION MATRIX

Permission Key	Super Admin	Admin	Sales Mgr	Sales Exec	Support	Buyer
manage_all	✓	—	—	—	—	—
manage_properties	✓	✓	—	—	—	—
manage_builders	✓	✓	—	—	—	—
manage_customers	✓	✓	✓	✓	—	—
manage_bookings	✓	✓	—	—	—	—
manage_booking_status	✓	—	✓	—	—	—
manage_leads	✓	—	✓	✓	—	—
manage_site_visits	✓	—	✓	✓	—	—
manage_revenue_rules	✓	—	—	—	—	—
manage_settlements	✓	✓	—	—	—	—
manage_reviews	✓	✓	—	—	✓	—
view_public_listings	✓	✓	✓	✓	✓	✓

Crucially, `manage_revenue_rules` is restricted exclusively to Super Admin, preventing unauthorized alteration of commission formulas.

D. Automated Booking-to-Revenue Engine

The core innovation of EstateFlow is its event-driven revenue distribution engine, triggered when a property booking advances to **Confirmed** or **Completed**.

$$C_{\text{raw}} = V_b \times (\omega_r / 100) \quad [\text{Percentage}] \mid \omega_r \quad [\text{Flat Fee}] \quad (1)$$

$$C_{\text{capped}} = \min(C_{\text{raw}}, \kappa_r) \quad (\kappa_r: \text{Rule Ceiling}) \quad (2)$$

$$T_{\text{withheld}} = C_{\text{capped}} \times 0.05 \quad (\text{Section 194H TDS}) \quad (3)$$

$$C_{\text{net}} = C_{\text{capped}} - T_{\text{withheld}} = C_{\text{capped}} \times 0.95 \quad (4)$$

ALGORITHM 1: AUTOMATED BOOKING-TO-REVENUE DISTRIBUTION WORKFLOW

```

1: Input: Database db, Booking b_id, Valuation V_b, Partner u_id, Type prop_type
2: Output: List of Generated Commission Records & Ledger IDs
3: existing ← Query CommissionRecord WHERE booking_id == b_id
4: if Count(existing) > 0 then return [] // Idempotency Guard
5: rules ← Query RevenueRule WHERE is_active == True ORDER BY priority DESC
6: for each rule in rules do
7:   if rule.min_val and V_b < rule.min_val then continue
8:   if rule.prop_type and rule.prop_type != prop_type then continue
9:   raw ← (rule.type == '%') ? (V_b * rule.val / 100) : rule.val
10:  if rule.cap and raw > rule.cap then raw ← rule.cap
11:  tds ← raw * 0.05 // 5% Statutory TDS Withholding
12:  net ← raw - tds
13:  Insert CommissionRecord(b_id, rule.id, raw, tds, net, CONFIRMED)
14:  wallet ← GetOrCreateWallet(user_id = u_id)
15:  bal_before ← wallet.balance
16:  wallet.balance ← wallet.balance + net; wallet.earned += net
17:  Insert WalletTransaction(wallet.id, CREDIT, net, bal_before, wallet.balance)
18:  Dispatch Notification(u_id, "Commission Credited", net)
19: end for
20: Commit db; return records

```

E. Conversion-Weighted Lead Scoring & Kanban Pipeline

Rather than processing leads chronologically, EstateFlow evaluates five weighted behavioral signals adapted from Venkatesh et al. [5]:

$$S_L = \min(100, \sum_{i=1..5} w_i \times I_i) \quad (5)$$

where weights represent: Site-Visit Confirmation (\$w_1 = 30\$), Budget-Inventory Match (\$w_2 = 25\$), Engagement Depth (\$w_3 = 20\$), Timeline Immediacy (\$w_4 = 15\$), and KYC Verification (\$w_5 = 10\$). Leads advance across six Kanban stages: New Lead → Contacted → Site Visit Scheduled → Negotiation → Booking Initiated → Closed Won/Lost.

V. RESULTS & DISCUSSION

A. Build & Compilation Verification

Static analysis on the React 19 codebase using `npx tsc --noEmit` completed with **0 errors** across all type definitions. Vite transformed 2,530 modules in **1.88 seconds**, yielding a core HTML payload of 1.62 kB (0.73 kB gzip) and base CSS of 70.17 kB (11.26 kB gzip), confirming lightweight client bundle characteristics.

B. API Latency & Throughput Profiling

Profiling the FastAPI ASGI server across 1,000 requests per route under Python 3.11 demonstrated compliance with the <40 ms industry SLA formulated by Venkatesh et al. [5]:

TABLE IV: API LATENCY PROFILING (1,000 SAMPLE QUERIES)

Endpoint Under Test	Method	Mean Latency	P95 Latency	P99 Latency
/api/health	GET	4.2 ms	6.8 ms	9.1 ms
/api/properties	GET	18.4 ms	26.5 ms	34.1 ms
/api/properties/{id}	GET	12.1 ms	18.0 ms	22.4 ms
/api/admin/dashboard/summary	GET	24.6 ms	38.2 ms	45.0 ms
/api/bookings (Revenue Trigger)	POST	34.2 ms	48.1 ms	56.3 ms

C. Automated End-to-End Staging QA Validation

An automated verification suite (run_staging_validation.py) evaluated complete customer and administrative workflows. All 16 phases achieved a **100% pass rate**.

TABLE V: AUTOMATED STAGING QA VALIDATION RESULTS

Phase	Subsystem Under Test	Expected Outcome	Status
P-01	FastAPI Health Check	HTTP 200 Healthy	PASS
P-02	Customer Registration	HTTP 200 Created	PASS
P-03	Customer Login & JWT	Valid Bearer Token Issued	PASS
P-04	Property Parametric Search	City & BHK Filter Verified	PASS
P-05	Property Details View (#1)	Payload with Builder Data	PASS
P-06	Customer Wishlist Addition	Record Persisted to DB	PASS
P-07	Book Physical Site Visit	Appointment Record Created	PASS
P-08	Submit Property Review	5-Star Review Persisted	PASS
P-09	Admin Login & Permissions	Role: super_admin Verified	PASS
P-10	RBAC Guard Interception	Customer Token Intercepted → 403	PASS
P-11	Admin KPI Aggregation	30 Properties, Verified Builders	PASS
P-12	Recharts Serialization	Valid Time-Series JSON	PASS
P-13	Create Property Listing (#31)	Luxury Villa Inserted	PASS
P-14	Update Price & Audit Log	PropertyStatusHistory Logged	PASS
P-15	Builder Verification CRM	RERA Verification State: Verified	PASS
P-16	Revenue Engine Execution	5% TDS Deducted + Wallet Credited	PASS

D. Discussion of Findings

Phase P-16 confirms that the revenue engine executes deterministically: updating a booking to Confirmed immediately withholds 5% TDS under Section 194H, credits the net balance to the channel partner's wallet, and logs an immutable `WalletTransaction` record. The sub-60ms P99 latency across all routes demonstrates the efficiency of FastAPI's ASGI event loop and SQLAlchemy eager relationship joins.

VI. CONCLUSION & FUTURE SCOPE

This project presented the design and development of EstateFlow CRM, a real estate CRM with an integrated Revenue Sharing Management System. By unifying client relationship tracking, behavioral lead scoring, and automated commission distribution within a high-concurrency cloud architecture (FastAPI + React 19), EstateFlow overcomes the fragmentation, commission disputes, and high failure rates of legacy systems. The platform delivers sub-60ms P99 latencies, enforces statutory 5% Section 194H TDS withholding, and achieved a 100% pass rate across 16 automated staging test suites.

Future extensions include: (i) integrating Ethereum/Polygon smart contracts for on-chain commission escrow, (ii) deploying collaborative filtering recommendation models for returning buyers following Venkatesh et al. [5], and (iii) incorporating conversational AI chatbots for real-time lead nurturing [6].

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